CUSTOMER PURCHASE BEHAVIOUR ANALYSIS

Retail Sales Analysis Using Python and RFM Customer Segmentation

PROJECT REPORT

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Abstract

This project analyses retail transaction data to understand sales patterns, product performance, geographic contribution, and customer purchasing behaviour. The work uses Python with Pandas for data preparation and analysis, and Matplotlib/Seaborn for visualisation. Recency, Frequency, and Monetary (RFM) measures are calculated to group customers into practical behavioural segments. The analysis is descriptive: it identifies patterns in the available transaction data and does not establish causal effects.

Keywords: retail analytics, customer behaviour, sales analysis, Python, RFM segmentation.

1. Introduction

Understanding customer purchasing behaviour helps retailers identify important products, monitor sales over time, compare geographic markets, and design more relevant customer engagement strategies. Transaction records provide evidence about what was purchased, when orders occurred, and how much was spent. This project uses an Online Retail-style transaction dataset to explore these questions.

2. Project Objectives

- Summarise the dataset and its main fields.
- Examine total sales, customers, products, and invoices.
- Identify the top products by recorded sales value.
- Visualise monthly sales and sales by country.
- Use RFM measures to create customer segments.
- Present findings and practical recommendations with clear limitations.

3. Dataset Description

The cleaned workbook used in the analysis contains 392,692 rows and 9 columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country, and Sales. These fields describe invoice identifiers, products, quantities, transaction dates, unit prices, customer identifiers, country, and recorded sales value.

Field	Description
InvoiceNo	Invoice or transaction identifier.
StockCode / Description	Product code and product name.
Quantity / UnitPrice	Units purchased and unit price.
InvoiceDate	Date and time associated with the invoice.
CustomerID	Customer identifier where available.
Country	Country associated with the transaction.
Sales	Recorded sales value for the row.

Important scope note: the available columns do not contain explicit sales channel (online/offline) or promotion/discount fields. Therefore, this report analyses the available retail transactions and does not claim to measure omnichannel behaviour or promotion impact.

4. Tools and Methodology

Tools: Python, Pandas, Matplotlib, Seaborn, and Excel. The analysis was conducted in Google Colab.

Workflow: load the cleaned workbook; inspect columns, data types, missing values, and duplicates; summarise sales and customer metrics; aggregate sales by product, month, and country; then calculate RFM measures and apply rule-based customer segmentation.

RFM definitions: Recency is the number of days since a customer's latest purchase relative to one day after the latest transaction date in the filtered data; Frequency is the number of distinct invoices per customer; Monetary is the sum of recorded sales value for that customer. For RFM calculations, rows with non-positive quantity or unit price and rows without CustomerID were excluded.

5. Overall Sales and Customer Metrics

Metric	Result
Rows in cleaned dataset	392,692
Columns	9
Total recorded sales	8,887,208.89
Unique customers	4,338
Unique products (StockCode)	Noted in notebook; verify before final submission if needed
Unique invoices	Noted in notebook; verify before final submission if needed
Average invoice value	479.56

The sales total is the sum of the Sales column in the cleaned workbook. Confirm how returns/cancellations were handled before interpreting this as final net sales. The dataset's currency is not stated in this report because it has not been independently confirmed.

6. Visualisations and Findings

Figure 1. Top 10 countries by recorded sales

[Insert the corresponding chart image here before final submission: sales_by_country.png]

The United Kingdom has the largest recorded sales contribution in the available data, followed by the Netherlands, EIRE, Germany, and France. This indicates that sales are concentrated in the UK within this dataset.

Figure 2. Monthly sales trend

[Insert the corresponding chart image here before final submission: monthly_sales_trend.png]

The monthly trend helps identify changes in recorded sales across the observation period. Peaks and declines should be interpreted alongside seasonality, incomplete periods, and data quality.

Figure 3. Top 10 products by recorded sales

[Insert the corresponding chart image here before final submission: top_10_products.png]

This chart highlights products contributing the most recorded sales value. High sales value does not necessarily mean the product has the highest unit volume or profit.

Figure 4. RFM-based customer segments

[Insert the corresponding chart image here before final submission: customer_segmentation.png]

RFM segmentation groups customers using recency, purchase frequency, and recorded monetary value. Segment labels are rule-based and should be treated as descriptive groups, not predictive classifications.

7. RFM Customer Segmentation

RFM analysis combines three customer-level indicators: recency, frequency, and monetary value. Each indicator was converted into a quintile-style score using ranked values. Lower recency in days means a more recent purchase, so the recency score was reversed: higher R_Score indicates a more recent customer. Frequency and monetary scores increase with the corresponding measure.

The project uses simple rules to assign customers to groups such as Loyal Active Customers, High Value Recent Customers, At Risk / Inactive Customers, New or Occasional Customers, and Regular Customers. These labels depend on the selected thresholds and are intended to help describe the customer base. They are not validated predictions of future behaviour.

8. Key Findings

- The cleaned dataset contains 392,692 transaction rows and 4,338 unique customers.
- Recorded sales total 8,887,208.89 in the dataset's unspecified currency; confirm treatment of returns and cancellations.
- The United Kingdom is the largest country by recorded sales (7,285,024.64), with a substantial gap to the next countries.
- The top-product chart provides a way to prioritise product-level review based on recorded sales value.
- Monthly sales visualisation supports trend and seasonality review.
- RFM segmentation creates a structured starting point for differentiating recent, frequent, high-value, and less-active customers.

9. Recommendations

- Review the strongest country market and investigate what drives its contribution, while considering that the dataset may not represent all markets equally.
- Monitor monthly sales and compare equivalent periods before making seasonal decisions.
- Use the top-product results to support inventory and merchandising reviews, alongside margin and stock availability data.
- Consider targeted re-engagement for the at-risk/inactive group and retention initiatives for loyal customers; test campaigns before assuming they will increase sales.
- For future work, add explicit channel, discount/promotion, cost/margin, and return/cancellation fields if omnichannel and promotion impact are project requirements.

10. Limitations

- The dataset does not provide explicit online/offline channel fields, so channel comparisons cannot be performed.
- The dataset does not provide explicit promotion or discount fields, so promotion impact cannot be measured.
- Sales totals may be affected by returns/cancellations depending on how the cleaned data was prepared.
- The currency was not independently confirmed.
- RFM segments use chosen rules and are descriptive rather than predictive.
- Results depend on the data quality and observation period of the supplied workbook.

11. Conclusion

This project demonstrates how retail transaction data can be prepared, summarised, visualised, and used for basic customer segmentation with Python. The analysis identifies geographic sales concentration, provides product and monthly sales views, and creates RFM-based customer groups. These findings can support further

business investigation, but decisions should consider the stated data limitations. In particular, the available dataset cannot support claims about online-versus-offline behaviour or the effect of promotions without additional fields or data.

Appendix: Project Files

- Cleaned dataset: Online_Retail_Cleaned_Preprocessed.xlsx
- Analysis notebook: Customer Purchase Behaviour Analysis (.ipynb)
- Visualisations: sales_by_country.png, monthly_sales_trend.png, top_10_products.png, customer_segmentation.png
- Customer-level output: Customer_Segmentation_Results.xlsx

Before submission, fill in the cover-page details, verify the currency and treatment of returns/cancellations, and ensure the four chart images are included.